

Probabilistic Geometry-based Manifold Calibration for Robust Point-Cloud Representation Learning

1 Overview of Supplementary Materials

This document serves as a supplementary resource to the main paper, offering a deeper investigation into our method, *i.e.*, Probabilistic Geometric Manifold Calibration (PGMC). It includes additional implementation details, comprehensive experimental results, extended discussions, and methodological clarifications that complement the primary text. This supplementary of the formal paper is structured as follows:

- Section 1: In this section, we provide further experimental details, including detailed statistics of all datasets used and an extended set of hyperparameter configurations applied during both training and testing.
- Section 2: Extensive explanation for our method.
- Section 3: A comprehensive understanding can be achieved with complete experimental results.
- Section 4: Additional qualitative analysis of our PGMC to show its generalizability and robustness.
- Section 5: Limitation discussions. We analyze the limitations of PGMC and highlight directions for future research.

Our code is at: <https://github.com/ConstantineWayne/PGMC>.

2 Additional Implementation Details

2.1 Dataset Details

As described in the formal paper, we adopt 3 multimodal point cloud datasets to evaluate our method. Here we provide more details of the datasets as follows:

(1) **ModelNet-C**: [5] is a corrupted point cloud dataset designed to evaluate the robustness of 3D point-cloud classification models. ModelNet-C is based on the widely used ModelNet40 dataset, integrating a set of synthetic yet realistic perturbations that emulate the degradations commonly observed in real-world 3D sensing. Unlike the original clean CAD models in ModelNet40, ModelNet-C introduces controlled corruptions to test how well a model can maintain performance under challenging conditions. Each corruption is provided at multiple severity levels (from 0-4). Among the corruptions introduced, there are 7 atomic corruptions, including adding global outliers, adding local outliers, dropping global structure, dropping local parts, rotation, scaling, and jittering. Other corruptions can be derived from these atomic corruptions. Following [6], in our implementation, we chose the intensity of 2, simulating the realistic implementation of 3D point cloud recognition models.

(2) **ScanObjectNN-C (SONN-C)**: [8] is built upon the ScanObjectNN dataset. Similar to ModelNet-C, we implemented the 7 atomic corruptions to simulate real-world scenarios. Additionally, SONN-C provides three benchmark variants that reflect different levels of real-world difficulty in 3D point-cloud recognition:

OBJ_ONLY, OBJ_BG, and HARDEST. (1) *OBJ_Only*: It contains objects that are cleanly segmented from real indoor scans. Although the point clouds preserve natural irregularities and sensor noise, there is no surrounding clutter. This variant evaluates a model's ability to recognize real objects without contextual distractions. (2) *OBJ_BG*: It introduces realistic background clutter by keeping the object within the original scene context. Here, the point clouds include surrounding structures, furniture, and environmental noise captured by RGB-D sensors. (3) *HARDEST*: This variant is the most challenging split and incorporates background clutter, occlusion, misalignment, and rigid transformations. Objects are not pre-aligned, and their orientations vary significantly across samples. HARDEST is designed to test robustness under severe distribution shift and imperfect sensing, reflecting the most realistic but demanding conditions for 3D recognition.

(3) **ShapeNet**: [1] is a dataset used for geometry understanding, shape analysis, and 3D deep learning research. The dataset is organized into multiple subsets, among which ShapeNet is the most commonly used. ShapeNet includes over 50,000 curated models spanning more than 50 semantic categories such as chairs, tables, airplanes, and vehicles. Each model is aligned, normalized, and annotated with rich metadata, including category labels, geometric attributes. Unlike real-world scanned datasets, ShapeNet offers clean geometry, free of sensor artifacts, occlusion, or background clutter. This allows controlled experimentation and serves as a benchmark for learning idealized 3D shape priors. At the same time, its scale and category diversity make it suitable for training robust models that can generalize across object families.

2.2 Task Implementation Details

Aside from the above seven point cloud corruptions, we also implemented typical adversarial attack to further test the adversarial robustness, using the ℓ_2 -constrained Fast Gradient Method (FGM- ℓ_2) and the Fast Gradient Sign Method (FGSM). For both attacks, we consider two attack intensity levels, denoted by $s = \{0.05, 0.1\}$.

Given an input point cloud $\mathbf{X}$ with its ground-truth label y , we compute the gradient of the classification loss with respect to the point coordinates as:

$$\mathbf{g} = \nabla_{\mathbf{X}} \mathcal{L}_{\text{cls}}(\mathbf{X}, y; \Theta), \quad (1)$$

where $\mathcal{L}$ is the Cross-Entropy loss, and Θ represents the parameters of the attacked model.

FGM- ℓ_2 generates an adversarial point cloud by perturbing the point coordinates along the normalized gradient direction:

$$\delta_{\text{FGM}}^{(s)} = \epsilon_2^{(s)} \frac{\mathbf{g}}{\|\mathbf{g}\|_2}, \quad \mathbf{X}_{\text{adv}}^{\text{FGM},(s)} = \mathbf{X} + \delta_{\text{FGM}}^{(s)}, \quad \epsilon_2^{(s)} = s \quad (2)$$

where ϵ_2 denotes the ℓ_2 perturbation budget.

FGSM attack is defined as:

$$\delta_{\text{FGSM}}^{(s)} = \epsilon_{\infty}^{(s)} \text{sign}(\mathbf{g}), \quad \mathbf{X}_{\text{adv}}^{\text{FGSM},(s)} = \mathbf{X} + \delta_{\text{FGSM}}^{(s)}, \quad \epsilon_{\infty}^{(s)} = s, \quad (3)$$

where ϵ_{∞} denotes the ℓ_{∞} perturbation budget.

Table 1: Quantitative comparison of 3D point-cloud classification accuracy (%) on the ModelNet-C dataset. We evaluate our PGMC against state-of-the-art methods [6, 7] based on diverse multimodal pre-trained models across seven atomic corruption.

Method	Clean	Corruption Type							Average
		Add Global	Add Local	Drop Global	Drop Local	Rotate	Scale	Jitter	
ULIP (CVPR'23)	56.16	35.55	43.92	54.70	50.89	55.27	50.20	44.08	47.80
Point Cache (CVPR'25)	64.22	46.15	47.98	59.16	56.00	61.47	55.35	49.92	53.72
Uni-Adapter (AAAI'26)	57.45	29.27	49.67	57.01	54.09	51.37	58.87	48.54	49.83
PGMC (Ours)	72.12	47.16	53.24	67.62	68.03	66.73	61.70	51.42	59.41
ULIP2 (CVPR'24)	71.23	65.15	54.62	68.76	57.98	70.30	67.10	21.76	57.95
Point Cache (CVPR'25)	74.53	68.11	61.62	73.22	63.65	73.34	70.42	29.50	62.84
Uni-Adapter (AAAI'26)	75.20	68.93	61.26	72.52	63.45	71.88	73.74	18.55	61.48
PGMC (Ours)	80.47	69.12	67.46	77.47	72.48	78.80	74.75	31.32	67.34
Openshape (NIPS'23)	84.56	71.64	67.79	81.56	73.58	82.01	78.48	59.36	73.49
Point Cache (CVPR'25)	84.52	74.84	73.70	82.41	76.26	83.18	78.93	68.35	76.81
Uni-Adapter (AAAI'26)	46.75	13.73	18.15	39.34	32.90	38.61	44.73	8.46	27.99
PGMC (Ours)	88.41	82.53	80.01	88.08	83.34	84.52	82.69	74.43	82.23
Uni3D (ICLR'24)	81.81	72.45	56.36	68.15	67.18	79.94	75.36	56.24	67.95
Point Cache (CVPR'25)	83.87	77.51	71.15	72.16	70.75	81.77	77.31	62.52	73.31
Uni-Adapter (AAAI'26)	85.69	67.54	68.84	73.58	72.28	79.66	83.00	62.96	72.55
PGMC (Ours)	88.24	78.36	73.01	79.99	75.64	83.58	80.23	77.83	78.38

Only the 3D point coordinates are perturbed, while the ground-truth labels remain unchanged. The resulting adversarial point clouds are then used as test-time inputs to evaluate the robustness.

For the simulation-to-real evaluation, we use ShapeNet as the synthetic source domain and three variants of ScanObjectNN-C as the real-world target domains: *Hardest*, *Obj_Only*, and *Obj_BG*. To ensure a consistent label space across the source and target domains, we retain the 9 categories shared by ShapeNet and ScanObjectNN-C: *bed*, *bag*, *cabinet*, *chair*, *display*, *pillow*, *shelf*, *table*, and *sofa*. Accordingly, we conduct three transfer evaluations: ShapeNet $\rightarrow$ ScanObjectNN-C (*Hardest*), ShapeNet $\rightarrow$ ScanObjectNN-C (*Obj_Only*), and ShapeNet $\rightarrow$ ScanObjectNN-C (*Obj_BG*). During test-time adaptation, the target-domain labels are not used and are accessed only to calculate the final accuracy.

2.3 Implementation Details

Instead of using a single rigid prompt such as “a point cloud object of a class” to describe each point cloud, we follow the strategy adopted in ULIP [9] and employ a set of 64 diverse text templates. Each template provides a slightly different linguistic description of the same 3D category, enhancing semantic richness and reducing prompt bias. These 64 descriptions are individually encoded into text embeddings, after which we take their average to obtain the final textual feature for each class. Moreover, we provide the detailed hyperparameter settings for our experiments on two tasks. For ModelNet-C and ScanObjectNN-C, we set the shot capacity K for positive memory in DCM and CCM in GPM to 25, and for negative memory to 10. For the calibration memory in SAM, we set the capacity in the DCM to 25, and for CCM to 10. For the sim2real task. We set the shot capacity K for positive memory in DCM and CCM in GPM to 25, and for negative memory to 10. For the calibration memory in SAM, we set the capacity in the DCM to 10, and for CCM to 5. For the construction of the f_θ in the CCM, we leveraged the attention-based network to aggregate the local

geometric information. We utilized the Adam to train the f_θ , the learning rate is set to $1e^{-4}$, while the weight decay is set to $1e^{-5}$. As for the test-time training batch of the two tasks, we set 128 for the Openshape and 64 for the rest of the pre-trained 3D models. Also, we set the training epoch to 15 for every task. For the inference, we set the batch size to 1 and report the average accuracy on the test set. As for the other hyperparameters, such as threshold γ , the lower bound is set to 0.2, while the upper bound is set to 0.5; and in positive memories for both calibration manifolds, α , β are set to 2 and 3, respectively, and in negative memories are set to 0.117 and 1. More implementation information, please refer to our code.

3 Theoretical Analysis of the MSD

As discussed in Section 3.1, we derive the Manifold Shift Degree (MSD) from the uncertainty mass of the subjective logic simplex. We first provide its formal definition. Given an input sample $\mathbf{x}$, let $\mathbf{L}(\mathbf{x}) = [\ell_1(\mathbf{x}), \dots, \ell_C(\mathbf{x})]$ denotes its classification logits over C semantic categories. The evidence associated with category c is obtained by applying the Softplus function:

$$e_c(\mathbf{x}) = \text{Softplus}(\ell_c(\mathbf{x})) = \log(1 + \exp(\ell_c(\mathbf{x}))). \quad (4)$$

The corresponding Dirichlet parameter and total Dirichlet strength are defined as:

$$\alpha_c(\mathbf{x}) = e_c(\mathbf{x}) + 1, \quad S(\mathbf{x}) = \sum_{c=1}^C \alpha_c(\mathbf{x}) = \sum_{c=1}^C e_c(\mathbf{x}) + C. \quad (5)$$

Following subjective logic, the belief mass assigned to category c and the uncertainty mass are given by:

$$b_c(\mathbf{x}) = \frac{e_c(\mathbf{x})}{S(\mathbf{x})}, \quad u(\mathbf{x}) = \frac{C}{S(\mathbf{x})}, \quad (6)$$

which satisfy $\sum_{c=1}^C b_c(\mathbf{x}) + u(\mathbf{x}) = 1$. We formally define the MSD:

$$\text{MSD}(\mathbf{x}) = \frac{C}{\sum_{c=1}^C e_c(\mathbf{x}) + C}. \quad (7)$$

Table 2: Comprehensive robustness evaluation on the ScanObjectNN-C dataset. The table reports classification accuracy (%) across three increasingly challenging benchmark subsets: *OBJ_BG*, *OBJ_Only*, and *HARDEST*.

Method	Clean	Corruption Type							Average
		Add Global	Add Local	Drop Global	Drop Local	Rotate	Scale	Jitter	
ScanObjectNN-C (OBJ_BG)									
ULIP (CVPR'23)	45.96	27.19	25.82	45.61	34.25	40.96	40.10	30.98	34.99
Point Cache (CVPR'25)	49.47	30.46	30.46	49.05	40.45	44.92	42.17	34.42	38.85
Uni-Adapter (AAAI'26)	49.74	30.63	32.53	52.32	43.54	39.75	40.97	31.67	38.77
PGMC (Ours)	65.06	34.59	41.82	63.51	60.41	58.86	58.17	39.93	51.04
ULIP2 (CVPR'24)	48.19	40.62	38.90	39.24	32.36	41.14	42.86	21.17	36.61
Point Cache (CVPR'25)	54.73	51.64	47.16	50.95	39.76	53.01	51.81	26.51	45.83
Uni-Adapter (AAAI'26)	50.77	41.65	41.13	43.54	31.84	46.29	52.15	15.66	39.63
PGMC (Ours)	64.54	50.95	54.56	57.48	50.94	61.27	54.73	36.14	52.30
Openshape (NIPS'23)	55.94	49.40	48.19	52.67	42.51	48.88	47.16	31.84	45.81
Point Cache (CVPR'25)	62.65	58.00	51.64	59.72	48.71	56.11	54.22	36.49	52.13
Uni-Adapter (AAAI'26)	59.03	56.11	49.57	55.25	44.40	48.02	48.70	33.21	48.24
PGMC (Ours)	67.81	60.75	62.99	67.98	56.28	62.99	62.65	44.75	59.77
Uni3D (ICLR'24)	60.24	58.00	52.32	51.64	44.23	58.00	51.81	39.24	50.75
Point Cache (CVPR'25)	63.86	66.27	57.83	58.52	50.77	61.62	59.55	46.30	57.27
Uni-Adapter (AAAI'26)	59.72	59.20	56.97	51.80	51.11	54.21	64.02	44.75	54.29
PGMC (Ours)	75.21	67.00	72.80	69.53	65.74	74.53	71.08	61.61	68.90
ScanObjectNN-C (OBJ_Only)									
ULIP (CVPR'23)	49.05	31.50	34.77	51.29	38.38	48.36	44.58	36.83	40.82
Point Cache (CVPR'25)	52.15	35.80	38.04	54.56	45.27	50.95	45.96	40.28	44.41
Uni-Adapter (AAAI'26)	55.93	33.73	35.80	57.07	54.09	45.26	51.11	35.11	44.60
PGMC (Ours)	62.65	39.75	45.61	65.06	57.48	59.89	54.90	50.43	53.30
ULIP2 (CVPR'24)	42.00	40.45	41.31	37.69	30.29	38.21	44.45	22.89	36.47
Point Cache (CVPR'25)	51.98	49.05	46.30	48.88	40.45	45.78	45.09	25.99	43.08
Uni-Adapter (AAAI'26)	50.25	41.99	37.00	45.09	40.10	53.35	45.43	24.78	41.11
PGMC (Ours)	61.10	47.00	57.65	59.72	53.18	62.30	58.00	39.75	53.94
Openshape (NIPS'23)	53.18	49.91	46.30	52.15	36.66	46.64	46.82	30.81	44.18
Point Cache (CVPR'25)	58.69	59.04	53.01	55.94	41.82	51.81	49.23	39.41	50.04
Uni-Adapter (AAAI'26)	52.15	48.19	45.61	50.77	35.97	44.06	44.06	37.34	43.71
PGMC (Ours)	65.92	59.55	56.45	58.86	48.70	57.14	58.00	48.02	55.25
Uni3D (ICLR'24)	65.58	62.65	56.45	60.07	49.40	61.62	56.11	43.55	55.69
Point Cache (CVPR'25)	70.22	65.40	59.38	64.20	54.91	63.34	62.13	53.18	60.36
Uni-Adapter (AAAI'26)	62.82	62.30	57.48	62.06	58.86	62.47	65.40	51.29	59.98
PGMC (Ours)	75.73	70.56	65.40	73.83	70.91	74.18	74.18	60.07	69.88
ScanObjectNN-C (Hardest)									
ULIP (CVPR'23)	29.29	19.26	18.39	30.99	23.91	27.48	26.34	21.44	23.97
Point Cache (CVPR'25)	32.48	23.46	22.69	34.70	31.75	33.00	28.63	25.05	28.47
Uni-Adapter (AAAI'26)	37.75	21.65	17.73	35.80	29.77	27.34	32.19	24.08	26.94
PGMC (Ours)	42.88	24.14	25.85	44.24	37.64	42.92	36.19	30.22	34.46
ULIP2 (CVPR'24)	33.38	30.29	29.42	28.24	24.91	28.56	30.22	12.98	26.37
Point Cache (CVPR'25)	42.40	35.70	34.42	37.75	34.21	36.26	36.09	19.12	33.36
Uni-Adapter (AAAI'26)	33.34	31.78	29.59	29.97	25.29	30.67	32.85	13.49	27.75
PGMC (Ours)	44.13	36.28	42.92	44.93	43.58	46.21	41.91	20.57	39.49
Openshape (NIPS'23)	41.12	32.41	35.60	37.80	27.34	36.61	35.22	18.88	31.98
Point Cache (CVPR'25)	43.72	40.91	39.24	43.03	35.22	43.06	38.24	25.05	37.82
Uni-Adapter (AAAI'26)	41.39	35.73	27.30	41.01	31.81	38.20	38.93	20.09	33.30
PGMC (Ours)	50.58	45.97	48.16	47.32	43.85	45.00	44.79	28.27	43.34
Uni3D (ICLR'24)	46.04	48.23	37.99	36.75	31.47	44.00	37.37	28.66	37.78
Point Cache (CVPR'25)	51.13	52.57	42.23	44.59	38.79	49.03	41.05	34.70	43.28
Uni-Adapter (AAAI'26)	44.37	47.12	37.61	39.93	38.58	33.10	45.45	41.11	41.33
PGMC (Ours)	66.03	55.03	50.00	55.27	54.54	62.21	54.58	39.90	53.08

Proposition 1. *The uncertainty u , derived from the augmented probability simplex, is inversely related to the local geometric discriminability of the semantic manifold $\mathcal{M}$.*

Proof. Let the total semantic evidence be: $E(\mathbf{x}) = \sum_{c=1}^C e_c(\mathbf{x})$. The MSD can then be expressed as:

$$u(\mathbf{x}) = \frac{C}{E(\mathbf{x}) + C}. \quad (8)$$

Its derivative with respect to the total evidence is:

$$\frac{\partial u}{\partial E} = -\frac{C}{(E + C)^2} < 0. \quad (9)$$

Therefore, the MSD is strictly decreasing with respect to the total amount of semantic evidence.

We next analyze its relationship with geometric discriminability. Let $k = \arg \max_i \ell_i$ denote the predicted category, and $r = \arg \max_{j \neq k} \ell_j$ denote the second-largest category. We measure local geometric discriminability using the logit margin:

$$m(\mathbf{L}) = \ell_k - \ell_r. \quad (10)$$

A larger margin indicates that the sample is more strongly aligned with the semantic anchor associated with category k and is more clearly separated from its closest competing anchor.

Consider a local variation directed by vertex:

$$\mathbf{L}(t) = \mathbf{L} + t\mathbf{q}_k, \quad (11)$$

where $\mathbf{q}_k$ is the k -th standard basis vector. This variation changes only the logit associated with the predicted semantic anchor, while all competing logits remain fixed. We restrict t to a local neighborhood in which the predicted and second-largest categories remain unchanged. The resulting margin is:

$$m(t) = \ell_k + t - \ell_r, \quad \frac{\partial m(t)}{\partial t} = 1. \quad (12)$$

Meanwhile, the MSD becomes:

$$u(t) = \frac{C}{\text{Softplus}(\ell_k + t) + \sum_{j \neq k} \text{Softplus}(\ell_j) + C}. \quad (13)$$

Since the derivative of the Softplus function is the sigmoid function, we obtain:

$$\frac{\partial u}{\partial m} = \frac{\partial u / \partial t}{\partial m / \partial t} = -\frac{C \sigma(\ell_k + t)}{\left(\text{Softplus}(\ell_k + t) + \sum_{j \neq k} \text{Softplus}(\ell_j) + C\right)^2} < 0. \quad (14)$$

Along a vertex-directed variation of the semantic manifold, increasing the logit margin decreases the MSD. This establishes the inverse relationship between the MSD and geometric discriminability.

The MSD therefore provides a scalar measure of the evidential support associated with a sample's alignment with the semantic manifold. A low MSD, i.e., $\text{MSD}(\mathbf{x}) \approx 0$, indicates strong accumulated semantic evidence. It also corresponds to a large logit margin and a stable alignment with the predicted semantic anchor. Such samples are consequently treated as reliable manifold anchors for constructing the calibration memory.

4 Additional Experiment Results

Due to the space limitation, we weren't able to conclude the complete experiment results on both point cloud recognition, cross-domain tasks, and adversarial attack robustness evaluation test to demonstrate the superiority of PGMC. Following the Point Cache [6], we use the four pre-trained multimodal 3D encoders, including ULIP [9], ULIP2 [10], Openshape [3], Uni3D [11], as the backbone and designed as a plug-and-play framework to enhance the robustness of pre-trained models through only test-time augmentation.

Complete Point-Cloud Recognition Evaluation. As summarized in Table 1 and Table 2, we evaluate PGMC against state-of-the-art methods across four pre-trained multimodal backbones. On ModelNet-C dataset, our proposed PGMC consistently delivers the highest average accuracy across all seven atomic corruptions. Notably, when integrated with the Openshape backbone, PGMC achieves an average accuracy of 82.23%, outperforming the baseline 73.49% and the competitive Point Cache 76.81% by significant margins. Furthermore, compared to the SOTA counterparts-Point Cache and Uni-Adapter, PGMC exhibits remarkable resilience against structural deficiencies, surpassing the former with 5.17 % enhancement in the face of point cloud corruption and the latter with 18.87 % on average in corruptive conditions. These results indicate that our probabilistic manifold calibration effectively reconstructs the semantic geometry even when severe occlusions occur, showing the superior robustness.

Moreover, according to Table 2, this superiority is further validated on the more challenging ScanObjectNN-C dataset, which introduces real-world sensing noise. Across all three subsets, our proposed PGMC demonstrates substantial performance gains. On the HARDEST split, PGMC elevates the Uni3D baseline from an average accuracy of 37.78% to an impressive 53.08%, which obtains 15.30% absolute improvement. Meanwhile, compared to Point Cache and Uni-Adapter, our PGMC still possesses the SOTA performances, gaining substantial enhancement in the face of corrupted point clouds. These comprehensive results on the ScanObject-NN datasets confirm that our method successfully preserves robust representations in real-world corrupted scenarios.

Complete Cross-Domain Evaluation. Bridging the synthetic-to-real gap remains a formidable challenge in 3D representation learning due to the intrinsic topological differences between different domains. Table 3 details the performance of our method on the cross-domain task. Our proposed PGMC consistently achieves the highest average accuracy across all combinations of backbones and dataset splits. For instance, in the challenging *Hardest* scenario using the Openshape backbone, PGMC secures an average accuracy of 54.00%, which effectively enhances the performance of the backbone and significantly surpasses Point Cache (45.81%) and Uni-Adapter (15.58%). Similar trends are observed across other configurations; for example, on the *Obj Only* split with the Uni3D backbone, PGMC reaches a peak accuracy of 65.89%, outperforming the state-of-the-art Point Cache by 2.56% and the Uni-Adapter by a substantial margin of 52.35%. These results demonstrate that while some methods like Uni-Adapter struggle significantly with the distribution shift in cross-domain tasks, our PGMC maintains robust cross-domain generalization and superior corruption resistance regardless of the underlying architectural backbone.

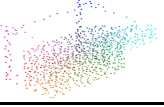
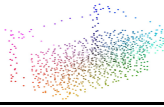
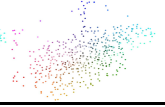
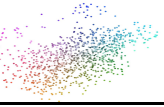
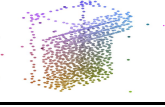
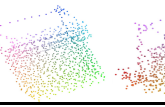
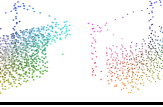
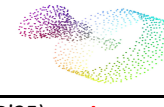
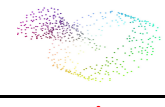
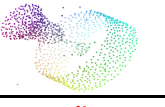
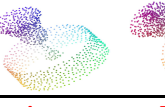
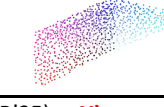
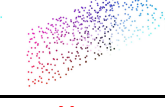
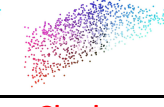
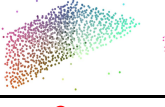
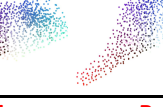
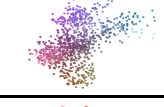
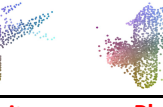
Clean	Scale	Drop Global	Jitter	Add Global	Rotate	Add Local	Drop Local
							
Point Cache (CVPR'25)	Sink	Sink	Xbox	Tvstand	Monitor	Range Hood	Sink
PGMC (Ours)	Bed	Bed	Bed	Bed	Bed	Bed	Bed
							
Point Cache (CVPR'25)	Plant	Desk	Tvstand	Radio	Plant	Radio	Radio
PGMC (Ours)	Lamp	Lamp	Lamp	Lamp	Lamp	Lamp	Lamp
							
Point Cache (CVPR'25)	Xbox	Vase	Glassbox	Cup	Car	Xbox	Bed
PGMC (Ours)	Laptop	Laptop	Laptop	Laptop	Laptop	Laptop	Laptop
							
Point Cache (CVPR'25)	Bottle	Bottle	Guitar	Tvstand	Tvstand	Guitar	Plant
PGMC (Ours)	Airplane	Airplane	Airplane	Airplane	Airplane	Airplane	Airplane

Figure 1: Qualitative analysis of our PGMC, compared to our counterpart method Point Cache, our method consistently produces the right prediction when it is in the face of different corruptions. Red and green text denote false and true predictions.

Complete Adversarial Attack Robustness Evaluation. Beyond natural corruptions, we investigate the vulnerability of point-cloud encoders to adversarial structural perturbations. As presented in Table 4, we subjected the models to FGM- l_2 and FGSM attacks under varying intensity levels ($\epsilon = 0.05, 0.1$). PGMC acts as a highly effective defensive mechanism across all datasets. Under the aggressive FGSM attack with $\epsilon = 0.05$ on ModelNet-C, PGMC integrated with Uni3D achieves an accuracy of 59.52%, significantly outperforming both the Point Cache and the recent Uni-Adapter with around 51.2% and 38.4% respectively. The margin of improvement is even more pronounced on ScanObjectNN-C splits. For instance, under the FGM- l_2 ($\epsilon = 0.1$) attack on the *Hardest* subset, PGMC consistently preserves a higher classification baseline. Our results demonstrate that PGMC not only excels in common corruptions but also serves as a generic and powerful defense, proving its generalizability and structural robustness in adversarial environments.

5 Additional Qualitative Analysis

Due to space limitations, we failed to include more qualitative analysis. In this supplementary material, we exhibit more cases selected from the ModelNet-C dataset to show the robustness of our model under different kinds of disturbances. The additional qualitative analysis is shown in the Fig. 1. We use the Point Cache [6], as our counterpart. It is obvious that our PGMC can consistently produce the correct predictions, regardless of the corruptions. While the Point Cache cannot make the right predictions due to the domain

shift. This additional qualitative analysis once again proves that our PGMC’s robustness in the face of various kinds of domain shift, *i.e.*, point cloud corruption, which are prevalent in realistic point cloud applications.

6 Limitation Discussion

In this work, we propose PGMC, a novel framework designed to improve the robustness of multimodal point cloud encoders under realistic corruptions during test-time inference. Specifically, PGMC is a manifold-calibration-driven framework that restores robust point-cloud representations by quantifying and correcting geometric distortion within a unified probability simplex. Central to this framework is the MSDQ module, which measures manifold shift through an augmented simplex and identifies reliable semantic anchors for calibration. Additionally, we develop a holistic calibration architecture that integrates Hybrid Calibration Memory for preserving ideal manifold geometry and Dual Manifold Calibration for dynamic, robust correction, jointly enabling effective recovery of degenerated manifolds. Extensive experiments conducted on two point cloud tasks (point cloud recognition and simulation-to-real) showed that our approach consistently achieves state-of-the-art performance and significant robustness to the domain shift.

Although PGMC demonstrates strong robustness, several limitations remain. First, the method assumes that corrupted manifolds still retain partial geometric cues that can be leveraged for

calibration; when distortions become extremely adversarial or non-structural, this assumption may weaken. Second, the selection of semantic anchors in MSDQ, while effective, may introduce bias when category boundaries are highly entangled or ambiguous. Additionally, the current calibration memory focuses primarily on geometric consistency, whereas other factors—such as feature uncertainty or cross-instance variability—are not explicitly modeled. Future work will investigate uncertainty-aware calibration and more adaptive anchor mechanisms to further enhance generalization.

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Table 3: Comparison with SOTA methods on cross-domain task. The dataset on the left side of $\rightarrow$ stands for simulated data, while the dataset on the right side indicates real-world data. Noting that Point Cache, Uni-Adapter, and our PGMC are based on the four backbones. The last column shows the average accuracy across all corruptions.

Method	Corruption Type							Avg
	Add Global	Add Local	Drop Global	Drop Local	Rotate	Scale	Jitter	
ShapNet → ScanObjectNN-C (Hardest)								
Ulip (CVPR'23)	46.19	46.24	45.34	37.48	39.79	38.58	26.44	40.00
Point Cache (CVPR'25)	45.39	47.15	45.74	39.04	42.97	38.53	24.58	40.48
Uni-Adapter (AAAI'26)	16.57	16.72	17.38	17.07	15.36	15.26	13.09	15.82
PGMC (Ours)	48.36	48.26	45.79	40.55	44.03	39.55	32.80	42.76
Ulip2 (CVPR'24)	40.95	40.20	34.15	29.67	33.04	37.48	16.62	33.15
Point Cache (CVPR'25)	47.30	47.41	43.77	40.10	42.57	40.45	24.18	40.82
Uni-Adapter (AAAI'26)	3.98	3.42	6.59	6.65	3.77	4.78	10.27	5.49
PGMC (Ours)	51.44	52.34	44.18	35.61	43.68	40.65	29.67	42.51
Openshape (NIPS'23)	51.99	57.73	49.31	36.92	48.11	43.97	20.15	44.02
Point Cache (CVPR'25)	51.23	52.94	49.57	48.46	49.26	42.57	26.70	45.81
Uni-Adapter (AAAI'26)	19.19	18.89	16.82	15.91	17.12	14.66	6.46	15.58
PGMC (Ours)	60.40	62.97	59.84	50.47	52.49	50.18	41.71	54.00
Uni3D (ICLR'24)	50.17	50.78	36.92	36.87	46.75	42.11	28.66	41.75
Point Cache (CVPR'25)	45.89	50.68	38.28	43.37	41.66	43.52	28.41	41.68
Uni-Adapter (AAAI'26)	13.90	13.80	11.78	10.42	11.48	13.84	9.01	12.55
PGMC (Ours)	58.49	57.27	47.10	43.68	59.95	47.61	41.06	50.73
ShapeNet → ScanObjectNN-C (Obj_Only)								
Ulip (CVPR'23)	49.50	60.75	60.50	53.00	57.75	56.50	45.25	54.75
Point Cache (CVPR'25)	60.50	59.00	61.50	55.75	55.75	55.25	39.00	55.25
Uni-Adapter (AAAI'26)	17.25	18.75	18.50	12.75	16.00	16.00	9.75	15.14
PGMC (Ours)	63.00	62.75	62.25	64.00	64.25	59.75	54.00	61.42
Ulip2 (CVPR'24)	49.50	49.25	46.75	37.25	43.00	46.50	29.75	43.14
Point Cache (CVPR'25)	56.50	57.00	52.25	50.75	50.75	53.50	31.75	43.10
Uni-Adapter (AAAI'26)	7.00	6.25	5.25	2.75	2.00	5.75	9.25	5.46
PGMC (Ours)	57.00	55.50	54.75	50.75	52.25	54.25	40.25	52.10
Openshape (NIPS'23)	63.50	61.50	58.50	47.25	60.00	57.50	35.25	54.78
Point Cache (CVPR'25)	60.00	61.50	64.75	49.50	57.25	61.25	37.50	55.96
Uni-Adapter (AAAI'26)	10.75	9.75	5.00	1.75	13.75	5.50	2.75	7.04
PGMC (Ours)	65.25	64.00	67.50	55.25	61.00	63.00	39.75	59.39
Uni3D (ICLR'24)	65.50	66.00	63.00	57.00	62.25	60.50	49.50	60.53
Point Cache (CVPR'25)	68.00	67.75	70.25	62.00	64.57	59.25	51.50	63.33
Uni-Adapter (AAAI'26)	15.50	14.00	15.75	13.25	11.25	15.75	9.25	13.54
PGMC (Ours)	72.00	70.50	67.75	64.25	64.25	65.00	53.50	65.89
ShapNet → ScanObjectNN-C (Obj_BG)								
Ulip (CVPR'23)	57.00	55.50	60.00	60.50	55.00	56.00	39.25	54.75
Point Cache (CVPR'25)	60.25	59.00	62.00	60.25	57.75	50.50	36.00	55.11
Uni-Adapter (AAAI'26)	14.50	15.25	18.50	16.50	15.75	14.50	13.50	15.50
PGMC (Ours)	68.00	67.00	66.75	62.75	63.75	58.75	45.25	61.75
Ulip2 (CVPR'24)	56.25	55.50	43.75	38.50	46.25	50.75	25.50	45.21
Point Cache (CVPR'25)	63.25	61.75	60.75	51.25	54.25	57.75	21.75	52.96
Uni-Adapter (AAAI'26)	10.25	10.50	9.50	6.25	6.75	3.25	9.75	8.04
PGMC (Ours)	67.25	65.75	55.00	54.50	60.75	59.50	34.25	56.71
Openshape (NIPS'23)	66.75	71.00	63.50	53.50	63.75	60.00	36.00	59.21
Point Cache (CVPR'25)	65.00	66.50	68.00	54.75	58.00	61.00	41.00	59.17
Uni-Adapter (AAAI'26)	15.00	14.50	9.50	6.25	17.75	14.25	7.25	12.07
PGMC (Ours)	67.00	70.50	70.50	55.00	68.00	61.25	45.25	62.50
Uni3D (ICLR'24)	67.50	67.50	51.75	47.25	64.25	56.50	42.75	56.78
Point Cache (CVPR'25)	65.00	71.00	57.50	57.50	64.25	52.25	44.75	58.89
Uni-Adapter (AAAI'26)	16.75	16.25	16.25	10.75	18.00	16.50	10.25	15.39
PGMC (Ours)	74.75	74.75	58.00	54.50	71.25	59.25	47.75	62.89

Table 4: Comparison under two adversarial attacks across four datasets (*ModelNet-C*, *Obj_Only*, *Obj_Bg*, and *Hardest*), i.e., FGM-l2 [4] and FGSM [2]. ϵ denotes the attack intensity in experiments.

Attack Methods	Method	$\epsilon = 0.05$				$\epsilon = 0.1$			
		ULIP	ULIP2	Openshape	Uni3D	ULIP	ULIP2	Openshape	Uni3D
ModelNet-C									
FGM-l2 [4]	Baseline	48.74	63.08	83.54	71.71	46.02	58.30	82.41	65.35
	Point Cache (CVPR'25) [6]	57.62	66.89	66.89	71.63	54.45	63.69	63.69	66.77
	Uni-Adapter (AAAI'26) [7]	52.18	67.42	41.93	80.38	50.00	64.26	40.55	76.78
	PGMC (Ours)	66.65	73.78	87.11	85.00	64.62	70.62	86.38	83.06
FGSM [2]	Baseline	41.20	44.89	47.93	43.84	34.84	33.22	31.62	34.44
	Point Cache (CVPR'25) [6]	44.24	44.52	44.52	46.39	37.03	31.72	31.72	33.26
	Uni-Adapter (AAAI'26) [7]	48.78	50.04	6.07	56.44	40.39	40.09	5.75	40.19
	PGMC (Ours)	49.55	53.07	56.11	59.52	42.22	40.35	37.43	40.96
ScanObjectNN-C (Obj_Only)									
FGM-l2 [4]	Baseline	27.02	26.16	50.08	37.34	19.44	22.54	47.67	32.35
	Point Cache (CVPR'25) [6]	34.59	31.32	53.87	41.48	31.32	28.91	50.08	35.62
	Uni-Adapter (AAAI'26) [7]	34.94	31.32	51.46	43.89	28.91	28.57	50.60	40.62
	PGMC (Ours)	55.42	49.91	65.23	60.24	51.11	45.95	64.02	52.49
FGSM [2]	Baseline	26.67	31.15	27.36	43.84	25.12	27.36	21.17	34.44
	Point Cache (CVPR'25) [6]	29.43	38.38	15.66	46.39	28.74	32.01	15.49	33.26
	Uni-Adapter (AAAI'26) [7]	29.43	31.32	32.01	38.72	35.97	22.54	11.53	33.20
	PGMC (Ours)	45.61	40.10	37.86	59.52	41.13	30.12	21.18	40.96
ScanObjectNN-C (Obj_Bg)									
FGM-l2 [4]	Baseline	22.37	26.85	51.11	32.35	15.49	25.30	48.36	25.47
	Point Cache (CVPR'25) [6]	32.18	34.25	57.65	37.17	30.80	31.84	53.52	31.32
	Uni-Adapter (AAAI'26) [7]	31.67	27.71	54.56	42.68	23.23	24.95	50.25	33.04
	PGMC (Ours)	55.07	49.05	66.09	57.83	47.84	43.02	64.71	50.08
FGSM [2]	Baseline	17.72	28.57	20.82	23.92	15.66	20.82	15.66	19.27
	Point Cache (CVPR'25) [6]	24.78	28.39	16.00	24.95	13.25	20.13	13.25	14.62
	Uni-Adapter (AAAI'26) [7]	25.81	29.77	20.99	31.66	20.18	22.20	9.98	22.20
	PGMC (Ours)	35.11	35.97	37.87	28.05	28.91	28.74	21.17	23.58
ScanObjectNN-C (Hardest)									
FGM-l2 [4]	Baseline	11.79	17.73	37.05	21.54	7.52	15.23	33.55	15.51
	Point Cache (CVPR'25) [6]	17.41	18.25	39.76	24.56	12.63	16.51	36.50	17.24
	Uni-Adapter (AAAI'26) [7]	16.55	16.82	40.87	27.62	12.08	15.71	37.40	22.20
	PGMC (Ours)	31.64	32.02	45.48	47.84	27.75	28.38	42.85	40.80
FGSM [2]	Baseline	12.56	18.32	15.82	15.85	12.56	15.12	11.90	12.80
	Point Cache (CVPR'25) [6]	15.12	19.53	14.71	15.30	13.42	14.85	10.61	13.32
	Uni-Adapter (AAAI'26) [7]	17.38	17.38	18.70	21.96	13.49	13.42	10.47	19.81
	PGMC (Ours)	25.85	23.62	25.19	25.98	24.21	20.09	15.30	19.63